

PROJECT REPORT

Employee Attrition Analysis & Prediction Using Machine Learning

1. Introduction

Employee attrition is one of the major challenges organizations face today. High employee turnover can reduce productivity, increase recruitment costs, and negatively affect organizational performance. Human Resource Analytics helps organizations understand the factors that influence employee retention and employee resignation.

This project applies Data Science and Machine Learning techniques to analyze employee attrition patterns and build predictive models capable of identifying employees likely to leave the company.

2. Objectives

The objectives of this project are:

- To understand and analyze the employee attrition dataset
- To clean and preprocess HR data
- To identify trends and factors affecting employee attrition
- To visualize employee-related insights
- To build predictive machine learning models for attrition prediction

3. Dataset Description

The dataset used in this project is the IBM HR Analytics Employee Attrition dataset. It contains employee-related information including demographic details, work experience, salary, job satisfaction, overtime, and attrition status.

Dataset Characteristics

- Total Records: 1470
- Total Features: 35
- Target Variable: Attrition

Important Features

- Age
- Department
- Gender
- Job Role
- Monthly Income
- Overtime
- Job Satisfaction

- Years at Company
- Attrition

4. Data Collection & Understanding

The dataset was loaded using the Pandas library. Initial analysis involved:

- Checking dataset dimensions
- Identifying data types
- Understanding feature meanings
- Reviewing statistical summaries

The target variable `Attrition` contained:

- Yes → Employee left the company
- No → Employee stayed in the company

For machine learning purposes:

- Yes was encoded as 1
- No was encoded as 0

5. Data Cleaning & Preprocessing

Data preprocessing was performed to ensure data quality and model readiness.

Steps Performed

- Checked missing values
- Removed duplicate records
- Encoded categorical variables
- Standardized numerical features

Tools Used

- LabelEncoder
- StandardScaler

The dataset was found to contain minimal missing data and was relatively clean.

6. Exploratory Data Analysis (EDA)

Exploratory analysis was conducted to identify trends, patterns, and relationships affecting employee attrition.

Key Analyses Performed

- Attrition distribution
- Attrition by department
- Attrition by gender
- Overtime impact analysis
- Salary distribution

- Job satisfaction analysis
- Correlation analysis

Major Findings

- Employees working overtime had significantly higher attrition rates.
- Employees with lower job satisfaction were more likely to leave.
- Lower monthly income contributed to higher turnover.
- Employees with fewer years at the company showed increased attrition.

7. Data Visualization

Several visualizations were created to communicate insights effectively.

Visualizations Included

- Count plots
- Pie charts
- Histograms
- Box plots
- Heatmaps

These visualizations helped identify patterns and relationships between employee characteristics and attrition behavior.

8. Machine Learning Modeling

Machine learning models were developed to predict employee attrition.

Models Used

1. Logistic Regression
2. Random Forest Classifier

Model Evaluation Metrics

- Accuracy Score
- Precision
- Recall
- F1-Score
- Confusion Matrix

The Random Forest model achieved stronger predictive performance compared to Logistic Regression.

9. Results

The predictive models successfully identified factors associated with employee attrition. Overtime, job satisfaction, monthly income, and years at company were among the most influential features.

The Random Forest model demonstrated high accuracy in predicting employee turnover.

10. Conclusion

This project demonstrated the application of Data Science techniques in Human Resource Analytics. Through data cleaning, exploratory analysis, visualization, and predictive modeling, valuable insights regarding employee attrition were identified.

The results can help organizations improve employee retention strategies and make data-driven HR decisions.

11. Future Improvements

Future enhancements may include:

- Hyperparameter tuning
- Deployment using Streamlit or Flask
- Advanced machine learning models
- Real-time employee attrition dashboard

12. References

- IBM HR Analytics Employee Attrition Dataset
- Scikit-learn Documentation
- Pandas Documentation
- Matplotlib Documentation
- Seaborn Documentation